

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 3

Case Study: Long-Run Price Elasticity in Brazilian Retail: An ADL and Error Correction Approach

Application area: *Marketing — Retail Pricing and Demand*

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IBGE Sistema Nacional de Índices de Preços ao Consumidor (<https://www.ibge.gov.br/estatisticas/economicas/precos-custos-e-indices-de-precos>): IPCA monthly price index by category. Alternatively, FRED BRACPIALLMINMEI for Brazil CPI and BRAGDPNADSMEI for GDP. For retail sales: PMC (Pesquisa Mensal de Comércio) from IBGE.

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1. Background and Objectives

Understanding the long-run price elasticity of retail demand is central to pricing strategy in marketing. If consumer price indices and retail sales volumes are both $I(1)$, a static OLS regression may be spurious. An error correction model (ECM) recovers both the short-run adjustment and the long-run elasticity.

Learning objectives: (i) Specify and estimate an ADL model; (ii) test for cointegration using the Engle–Granger two-step procedure; (iii) reparameterise as an ECM; (iv) interpret short-run and long-run price elasticities; (v) assess weak exogeneity of prices.

2. Data Preparation

```
library(dynlm); library(urca); library(tseries)
# Monthly retail volume index (PMC) and IPCA from IBGE
# Load as zoo or ts objects; log-transform both series
lsales <- log(retail_volume_index) # log retail sales
lprice <- log(IPCA_index)          # log price level
```

3. Exercises

3.1. Unit root pre-testing

Apply ADF and KPSS tests to `lsales` and `lprice`. Confirm that both are $I(1)$.

3.2. Estimating the ADL model

Estimate ADL(2,2) of log sales on log prices:

$$\text{lsales}_t = \alpha_0 + \alpha_1 \text{lsales}_{t-1} + \alpha_2 \text{lsales}_{t-2} + \beta_0 \text{lprice}_t + \beta_1 \text{lprice}_{t-1} + \beta_2 \text{lprice}_{t-2} + \varepsilon_t.$$

- Report estimates and standard errors.
- Compute the long-run price elasticity $\theta = (\beta_0 + \beta_1 + \beta_2)/(1 - \alpha_1 - \alpha_2)$.
- Is the long-run elasticity negative (as theory predicts)?

3.3. Engle–Granger cointegration test

- (a) Estimate the static long-run regression: $\text{lsales}_t = \mu + \theta \text{lprice}_t + \hat{u}_t$.
- (b) Apply the ADF test to $\hat{u}_t$ using MacKinnon (1991) critical values. Do you reject non-cointegration?
- (c) Interpret: what does cointegration mean economically for the sales–price relationship?

3.4. ECM reparameterisation

Reparameterise your ADL(2,2) as an ECM:

$$\Delta \text{lsales}_t = \tilde{\alpha}_1 \Delta \text{lsales}_{t-1} + \beta_0 \Delta \text{lprice}_t + \tilde{\beta}_1 \Delta \text{lprice}_{t-1} - \pi [\text{lsales}_{t-1} - \hat{\theta} \text{lprice}_{t-1}] + \varepsilon_t.$$

- (a) Is $\hat{\pi} < 0$? Is it statistically significant?
- (b) Interpret the speed of adjustment: how many months does it take to correct half the deviation from equilibrium?
- (c) Compare the short-run and long-run price elasticities. Which is larger in absolute value?

3.5. Weak exogeneity test

Estimate a second ECM with Δlprice_t on the left-hand side and the same ECT on the right-hand side. Is the ECT loading coefficient significant? What does this imply about the weak exogeneity of prices for the long-run parameter θ ?

If prices are weakly exogenous, the ECT loading in the price equation should be statistically zero. This would validate the single-equation ECM for sales as an efficient estimator of the long-run elasticity.

4. Core References

Engle, R.F. & Granger, C.W.J. (1987). *Econometrica*, 55, 251–276. Engle, R.F., Hendry, D.F. & Richard, J.-F. (1983). *Econometrica*, 51, 277–304. Dekimpe, M.G. & Hanssens, D.M. (1995). *Marketing Science*, 14, 1–21.

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